MAGIVOX AI

Java Programming - A to Z

Core Java | OOP | Collections | Exceptions | JDBC | Interview

How to use this guide
Study one topic, type the example yourself, practise without copying, then take the MAGIVOX topic test. After all modules, take the final course exam.

Learning outcomes

- Java Foundation
- Variables and Data Types
- Operators and Control Flow
- Methods and Arrays
- Object-Oriented Programming
- Strings and Wrappers

Table of Contents

1. Java Foundation
2. Variables and Data Types
3. Operators and Control Flow
4. Methods and Arrays
5. Object-Oriented Programming
6. Strings and Wrappers
7. Collections Framework
8. Exception Handling
9. File Handling
10. Modern Java Basics
11. JDBC Basics
12. Testing and Debugging
13. Mini Project
14. Interview Preparation

1. Java Foundation

What to learn

- JDK, JRE and JVM
- How Java source becomes bytecode
- Install JDK and run javac/java
- Identifiers, keywords and comments

Practice checkpoint

Explain jdk, jre and jvm in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

2. Variables and Data Types

What to learn

- Primitive types and ranges
- Reference types
- Type casting
- final constants
- Scanner input and formatted output

Practice checkpoint

Explain primitive types and ranges in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

3. Operators and Control Flow

What to learn

- Arithmetic, relational and logical operators
- if/else and switch
- for, while and do-while
- break and continue
- Nested loops

Practice checkpoint

Explain arithmetic, relational and logical operators in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

4. Methods and Arrays

What to learn

- Method signature and return type
- Parameters and method overloading
- One-dimensional and multidimensional arrays

- Arrays utility methods

Practice checkpoint

Explain method signature and return type in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

5. Object-Oriented Programming

What to learn

- Class and object
- Constructors
- Encapsulation
- Inheritance
- Polymorphism
- Abstraction
- Interfaces

Practice checkpoint

Explain class and object in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

6. Strings and Wrappers

What to learn

- String immutability
- StringBuilder
- Important String methods
- Wrapper classes and autoboxing

Practice checkpoint

Explain string immutability in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

7. Collections Framework

What to learn

- List, Set and Map
- ArrayList vs LinkedList
- HashSet and TreeSet
- HashMap and TreeMap
- Iterator and enhanced for loop
- Comparable vs Comparator

Practice checkpoint

Explain list, set and map in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

8. Exception Handling

What to learn

- Checked vs unchecked exceptions
- try/catch/finally
- throw and throws
- Custom exceptions

Practice checkpoint

Explain checked vs unchecked exceptions in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

9. File Handling

What to learn

- Path and Files
- BufferedReader/Writer
- Try-with-resources
- CSV-style text processing

Practice checkpoint

Explain path and files in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

10. Modern Java Basics

What to learn

- Lambda expressions
- Functional interfaces
- Streams
- Optional
- Date and Time API

Practice checkpoint

Explain lambda expressions in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

11. JDBC Basics

What to learn

- Driver and connection
- PreparedStatement

- SELECT/INSERT/UPDATE
- ResultSet
- Transactions and resource cleanup

Practice checkpoint

Explain driver and connection in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

12. Testing and Debugging

What to learn

- Read stack traces
- Common NullPointerException causes
- Unit-test mindset
- Debug checklist

Practice checkpoint

Explain read stack traces in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

13. Mini Project

What to learn

- Build a console Employee/Student manager
- Add CRUD operations
- Validate input
- Store records in file or database

Practice checkpoint

Explain build a console employee/student manager in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

14. Interview Preparation

What to learn

- OOP pillars
- == vs equals
- String vs StringBuilder
- ArrayList vs LinkedList
- HashMap working idea
- Exception hierarchy
- final/finally/finalize distinction

Practice checkpoint

Explain oop pillars in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

Worked Examples

Method example

```
static int add(int a, int b) { return a + b; }
```

OOP example

```
class Employee { private String name; Employee(String name){ this.name=name; } String getName(){ return name; } }
```

Collection example

```
Map<String,Integer> scores = new HashMap<>(); scores.put("SQL", 85);
```

Interview Answer Pattern

Use this structure for technical questions:

- 1. Definition - say what it is in one or two lines.
- 2. Why - explain why it is useful.
- 3. Example - give a small real or code example.
- 4. Difference/common mistake - compare with a related concept when relevant.
- 5. Be truthful - do not claim tools or experience you have not used.

Final Course Checklist

- I can explain the foundation without notes.
- I completed examples by typing/doing them myself.
- I completed topic tests and reviewed wrong answers.
- I completed at least one mini project or practical exercise.
- I can answer common interview questions in simple language.
- I am ready for the MAGIVOX 50-question final exam.